

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

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Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This project addresses the gap between theoretical schedulability and runtime deadline misses caused by memory contention in embedded real-time systems, where timing determinism is the critical concern. The proposed solution is a lightweight user-space FreeRTOS wrapper that extends EDF with a pairwise memory penalty model and a dynamic slack check, deferring high-memory tasks only when slack permits and falling back to pure EDF otherwise. Its main strengths are the non-invasive design, $O(n)$ overhead, and the hard safety guarantee; the main limitation is the simplified linear penalty model, which requires offline calibration of α and may not reflect real cache behavior. The most challenging component is accurate memory intensity profiling at design time. The evaluation metrics (DMR, jitter, contention penalty) are appropriate, but testing with only three tasks leaves scalability unclear.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

2.5D chiplet networks suffer from deadlocks due to cyclic inter-chiplet routing and from reliability failures caused by faulty vertical links, making fault tolerance and reliability the key concerns. DeFT addresses this with 3 virtual network rules that guarantee deadlock freedom and LUT-based fault-aware routing at boundary routers, evaluated on a cycle-accurate Noxim simulator. The strengths are the tight resource budget (2 VCs, 4-flit buffers) and 100% reachability guarantee under up to 25% fault rate. The main limitation is that this is a replication study, so novelty is limited. The most critical component is the LUT mechanism, which may not scale well with more chiplets. The evaluation plan is appropriate, but adding MTR or RC as baselines would make the comparison more informative.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project addresses real-time decoding of dense vehicle CAN traffic on a constrained MCU, where timing and reliability are the key concerns given frames arriving every 250–300 μ s. The approach is a four-stage pipeline (acquisition, buffering, processing, output) on an STM32F411 with Zephyr RTOS, using two-stage buffering to decouple fast reception from slow SD writes. The strengths are the clear timing budget and the passive monitoring design that is safe for vehicle networks. The main limitation is the lack of a DBC file; signals are inferred experimentally, making the decoder vehicle-specific. The most critical component is shared SPI arbitration between the CAN controller and SD card. The evaluation scenarios are appropriate, but no implementation results are shown yet.

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

SmartAquaria solves the problem of hardware-coupled embedded code that is difficult to port and test, using an aquarium life-support controller as the application domain, where reliability and energy efficiency are the key concerns. The approach is a strictly layered C++ architecture with dependency inversion: domain logic only knows abstract interfaces, enabling the same binary to run on real hardware or full software simulation. A single-threaded event FSM eliminates data races without locks. The strengths are clean portability, hardware-free testing with zero production delta, and a memory-efficient 512B tile renderer. The main weakness is that the feeding actuator remains mocked and no physical hardware results are presented. The most critical component is maintaining strict layer boundaries throughout the codebase. The evaluation is simulation-only; physical deployment results and energy measurements are missing.

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project addresses survivor localization in post-collapse rubble, where survival probability drops sharply after 72 hours and traditional methods fail. Key embedded constraints include microsecond clock synchronization ($<10\ \mu\text{s}$ drift via GNSS PPS), 24+ hour battery life via deep-sleep duty cycling, and LoRa's strict 48-byte payload limit. Four ESP32-S3 edge nodes capture audio, run a TinyML classifier to filter false positives, and transmit GNSS-timestamped events to a Jetson Nano that solves coordinates via a four-stage TDoA pipeline. The main limitation is that the star topology is prone to collisions in dense deployments. The GNSS PPS module is the most critical component, and the evaluation plan is well-defined with concrete targets for latency, synchronization drift, PDR, and localization accuracy.

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project targets vestibular dysfunction causing falls in elderly patients — a critical safety problem where existing GVS systems are open-loop and lab-bound. The STM32F411 running FreeRTOS handles a 100 Hz closed-loop pipeline: MPU6050 IMU $\rightarrow$ complementary+Kalman fusion $\rightarrow$ discrete PID $\rightarrow$ Howland current pump delivering 0–2 mA to mastoid electrodes. Key embedded constraints include a hard 15 ms E2E latency, 128 KB RAM, and a strict 2 mA safety limit enforced by three independent layers (HW comparator, RTOS monitor, watchdog). RMA analysis confirms schedulability ($U=0.53$). The main limitation is that evaluation is planned on a mechanical platform, not human subjects. The Howland current pump is the most critical component as it must deliver load-independent current regardless of electrode impedance changes. The evaluation plan is thorough, with concrete metrics for sway reduction, latency, jitter, and fault injection tests.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

This project addresses the limitation of static PID controllers in quadrotors, which fail to remain optimal under varying noise and environmental conditions. The NVIDIA Jetson platform is chosen for its heterogeneous CPU+GPU architecture and shared DRAM, eliminating PCIe transfer latency — a key embedded design consideration. The ARM CPU runs a 5 ms real-time control loop, while the CUDA GPU concurrently evaluates hundreds of PID gain candidates via Monte Carlo rollouts (GATE framework), and the best gains are fed back mid-flight. The main strength is the novel use of GPU parallelism for self-tuning on an embedded device; the limitation is that the system is simulation-only, with no real drone tested. The shared memory is the most critical component, as CPU and GPU compete for the same DRAM. Hard constraints (>1000 MC simulations/s, <15s tuning deadline, <2KB per thread) provide a concrete and sufficient evaluation plan.

Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

This project targets early fault detection in BLDC motors to prevent costly industrial downtime. The system uses an STM32WLE5 with a MEMS accelerometer, ultrasonic microphone, and NTC thermistor, transmitting 18-byte feature vectors over LoRaWAN to a fog-layer ML classifier. A condition-aware Random Forest (load and speed as features) achieves 92% file-level accuracy; the biggest gain came from merging indistinguishable fault classes rather than algorithm tuning. The main limitation is that results are based on a public dataset — real field validation with their custom PCB is still pending. The STM32WLE5 is the most critical component, and the evaluation plan is thorough, though hardware testing remains as future work.

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

Mapping applications to NoC cores is NP-Hard, and poor placement causes high latency and thermal hotspots. The proposed approach uses spectral clustering (Fiedler vector, silhouette-based K selection) to generate elite seeds for 20% of the NSGA-II initial population, optimizing communication cost and thermal balance simultaneously. The main strength is that spectral seeding collapses the search space early, likely accelerating convergence. The main limitation is the XY routing assumption, which ignores real contention effects. The most critical component is the Fiedler vector clustering quality, since poor seeds would reduce the benefit over standard NSGA-II. The evaluation uses standard benchmarks and appropriate baselines, but no results are shown yet; adding a Hypervolume indicator would better quantify Pareto front quality.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

This project tackles obstacle avoidance for autonomous miniature vehicles, a relevant problem for safe indoor navigation. The system uses a Raspberry Pi 3 with an RPLidar A2 sensor and L293D motor driver, where real-time LIDAR processing latency is the main embedded constraint. The LIDAR-only approach (no computer vision) enables low-light operation, and a five-stage reactive algorithm handles localization and decision-making. A notable addition is hardware emulation via Renode for pre-deployment testing. The main limitation is that the system is restricted to indoor, static environments. The RPLidar A2 is the most critical component, and while physical experiments are planned, the evaluation lacks specific metrics or test scenarios.